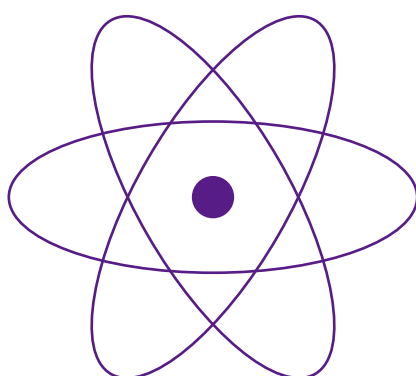


SPhOH

SAUDI PHYSICS OLYMPIAD HUB



Trigonometry I

Olympiad Problem Set

Olympiad Training • Math

SPhOH Team

FIRST EDITION • 2026

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About This Problem Set

This problem set is part of the training resources of the **Saudi Physics Olympiad Hub (SPhOH)**.

It is designed primarily for students preparing for the Saudi Physics Olympiad and related advanced physics competitions. The purpose of the handout is to provide a focused collection of problems that develops trigonometric fluency and its application to physical problems.

The handout is **problem-centered**. Explanations are intentionally brief. The main purpose is to provide deliberate problem-solving practice.

How to Use the Problem Set

Students are encouraged to attempt each problem independently before looking at its hint. A hint should be treated as a way to overcome a specific obstacle, not as a substitute for solving the problem.

For some problems, multiple hints may be provided. Try to use the least amount of assistance necessary before returning to the problem.

Difficulty Levels

The normal problems are divided into three training levels:

- **Level I** — Fundamental Olympiad problem-solving practice.
- **Level II** — Problems requiring combinations of ideas.
- **Level III** — More serious problems that require deeper analysis.

Find more at: <https://devfhd.github.io/SPhOH/>.

Trigonometry I

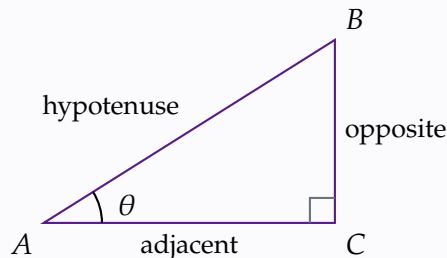
Topic Note.

Trigonometry provides a language for describing angles, triangles, directions, and components. In physics, the same ideas appear naturally in vectors, forces, motion, geometry, and many other areas.

Here are some important definitions:

1. Right-Triangle Trigonometry

For a right triangle ABC where $C = 90^\circ$:



- **Pythagorean Theorem:**

$$AC^2 + BC^2 = AB^2$$

- **Sine:**

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

- **Cosine:**

$$\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$$

- **Tangent:**

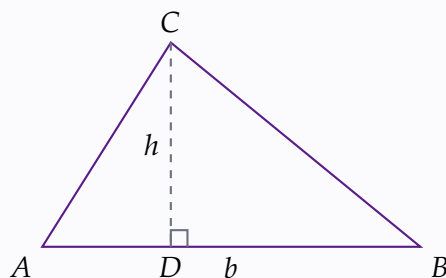
$$\tan \theta = \frac{\text{opposite}}{\text{adjacent}} = \frac{\sin \theta}{\cos \theta}$$

2. Area of a General Triangle

The area of any triangle is equal to half the product of a chosen base and its corresponding perpendicular height:

$$A = \frac{1}{2}bh$$

Here, b is the length of the base and h is the perpendicular distance from the opposite vertex to the line containing the base. The height is always perpendicular to the base; it is not necessarily one of the sides of the triangle.



In the diagram, $AB = b$ is the base and $CD = h$ is the corresponding height, with $CD \perp AB$. Therefore,

$$A_{\triangle ABC} = \frac{1}{2}(AB)(CD) = \boxed{\frac{1}{2}bh}.$$

The formula applies to every triangle, including acute, right, and obtuse triangles, provided the height is measured perpendicular to the chosen base.

3. Triangle Inequality

The side lengths of a triangle must satisfy the triangle inequality: each side must be strictly greater than the difference between the other two sides and strictly less than their sum.

For a triangle with side lengths a , b , and c :

$$\boxed{|b - c| < a < b + c}$$

Similarly,

$$\boxed{|a - c| < b < a + c}$$

and

$$\boxed{|a - b| < c < a + b}.$$

These inequalities can also be written as

$$a + b > c,$$

$$a + c > b,$$

$$b + c > a.$$

Thus, for three positive lengths to form a non-degenerate triangle, the sum of any two sides must be greater than the third side.

For example, lengths 3, 4, and 6 can form a triangle because $3 + 4 > 6$. However, lengths 2, 3, and 6 cannot, because $2 + 3 < 6$.

Equality is not allowed: if the sum of two sides equals the third, the points lie on a straight line and do not form a non-degenerate triangle.

1.1 Trigonometric Identities

PROBLEM 1

LEVEL I

The Fundamental Identity

Show that

$$\sin^2 x + \cos^2 x = 1.$$

PROBLEM 2

LEVEL II

A Tangent Identity

Show that

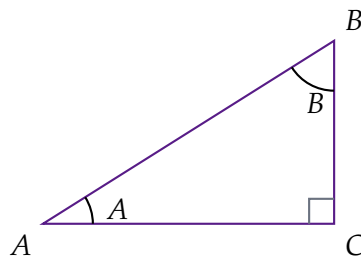
$$1 + \tan^2 x = \frac{1}{\cos^2 x}.$$

PROBLEM 3

LEVEL II

Complementary Angles

A right triangle ABC has $C = 90^\circ$.



(a) Show that

$$A + B = 90^\circ.$$

(b) Show that

$$\cos B = \sin A.$$

(c) Show that

$$\cos B = \sin(90^\circ - B).$$

(d) Explain geometrically why the two expressions are equal.

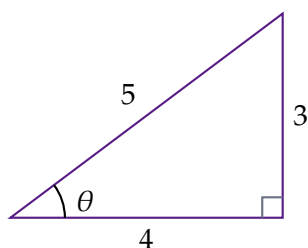
1.2 Right-Triangle Trigonometry

PROBLEM 4

LEVEL I

The 3–4–5 Triangle

Without using a calculator, find all six trigonometric ratios for a right triangle whose sides are 3, 4, and 5.



PROBLEM 5

LEVEL I

A Right Triangle with Known Sides

Triangle ABC is right-angled at C . The hypotenuse is 16 cm, and $AC = 12$ cm.

- (a) Find BC .
- (b) Find $\sin B$.
- (c) Find $\cos B$.
- (d) Find $\tan C$.
- (e) Explain why $\tan C$ has that result.

PROBLEM 6

LEVEL II

Working Backwards from Sine

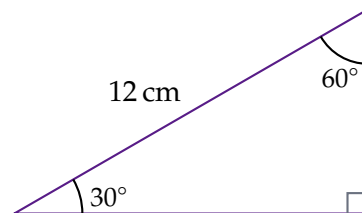
Triangle XYZ is right-angled at Z. $XY = 10$ cm and

$$\sin X = \frac{3}{5}.$$

- (a) Find XZ.
- (b) Find YZ.
- (c) Find $\cos X$.
- (d) Find $\tan Y$.

PROBLEM 7**LEVEL II****The 30–60–90 Triangle**

A right triangle has one acute angle of 30° and a hypotenuse of 12 cm.



- (a) Find the two shorter sides.
- (b) Find the area of the triangle.
- (c) Explain why a 30° - 60° - 90° triangle always has the same side ratio.

PROBLEM 8**LEVEL I****Exact Trigonometric Values**

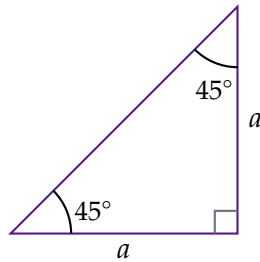
Without using a calculator, find:

- (a) $\sin 30^\circ$
- (b) $\cos 60^\circ$

- (c) $\sin 45^\circ$
- (d) $\cos 45^\circ$
- (e) $\tan 45^\circ$

PROBLEM 9**LEVEL II****The Right Isosceles Triangle**

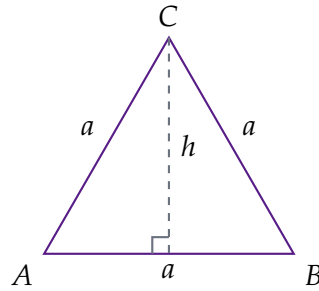
A right isosceles triangle has two equal legs of length a .



- (a) Show that its hypotenuse is $a\sqrt{2}$.
- (b) Determine its two acute angles.
- (c) Find $\sin 45^\circ$, $\cos 45^\circ$, and $\tan 45^\circ$.

1.3 Special Triangles and Geometry**PROBLEM 10****LEVEL II****Area of an Equilateral Triangle**

An equilateral triangle has side length a .
Derive an expression for its area in terms of a .

**PROBLEM 11**

LEVEL II

Dividing an Equilateral Triangle

An equilateral triangle has side length 10 cm.

- (a) Find its height.
- (b) Find its area.
- (c) Find the sine and cosine of the 30° and 60° angles formed when the triangle is divided into two right triangles.

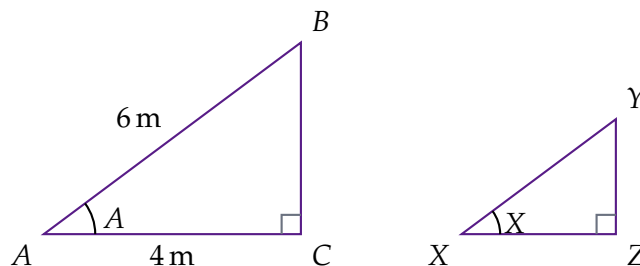
1.4 Similarity and Trigonometric Ratios

PROBLEM 12

LEVEL II

Similar Right Triangles

Two similar right triangles ABC and XYZ are shown. C and Z are right angles. $AB = 6$ m and $AC = 4$ m.



- (a) Find BC .
- (b) Find $\cos A$.
- (c) Find $\cos X$ if $A = X$.
- (d) Explain why the size of a triangle does not affect its trigonometric ratios.

PROBLEM 13

LEVEL II

Same Angle, Different Size

Two right triangles have the same acute angle θ but different side lengths.

Triangle 1 has opposite and adjacent sides of 3 cm and 4 cm.

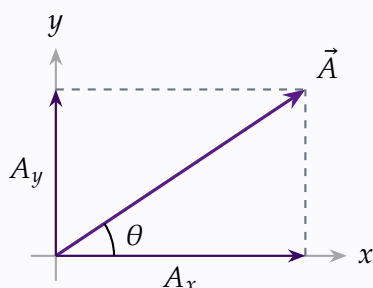
Triangle 2 has an adjacent side of 10 cm.

- (a) Find the opposite side of Triangle 2.
- (b) Explain why you can determine it without finding θ .

1.5 Trigonometry in Physics

In physics, vectors are often resolved into perpendicular components. Trigonometry provides the connection between the magnitude of a vector, its direction, and its components.

For Vector A , where θ is taken from positive x axis:



- **Pythagorean Theorem:**

$$A_x^2 + A_y^2 = A^2$$

- **Components:**

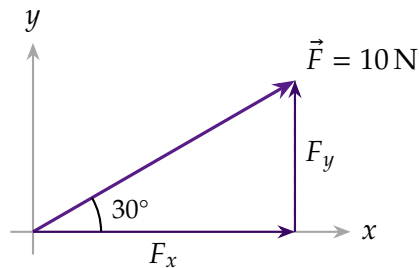
$$A_y = A \sin \theta, A_x = A \cos \theta$$

- **Angle Identity:**

$$\theta = \tan^{-1}\left(\frac{A_y}{A_x}\right)$$

PROBLEM 14**LEVEL II****Resolving a Force**

A 10 N force acts at an angle of 30° above the horizontal.



- Find the horizontal component of the force.
- Find the vertical component.
- Show that the original force can be reconstructed from its two components.

PROBLEM 15**LEVEL II****Velocity Components**

A vector has a magnitude of 20 m/s and makes an angle of 60° with the horizontal.

- Find its horizontal component.
- Find its vertical component.
- Which component is larger, and why?

PROBLEM 16**LEVEL II****Recovering an Unknown Direction**

A 50 N force acts at an unknown angle θ . Its horizontal component is 30 N.

- (a) Write an equation involving $\cos \theta$.
- (b) Find θ .
- (c) Find the vertical component.

PROBLEM 17

LEVEL II

Velocity of a Ball

A ball is moving with a velocity of 12 m/s at an angle of 40° above the horizontal.

- (a) Find v_x .
- (b) Find v_y .
- (c) Explain why the horizontal and vertical components are not both 12 m/s.

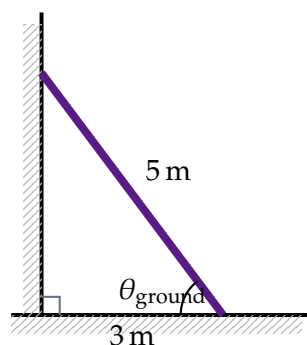
1.6 Geometric Applications

PROBLEM 18

LEVEL II

A Ladder Against a Wall

A ladder of length 5 m leans against a vertical wall. Its base is 3 m from the wall.



- (a) Find the height reached by the ladder.

- (b) Find the angle between the ladder and the ground.
- (c) Find the angle between the ladder and the wall.

PROBLEM 19**LEVEL II****A Sloping Ramp**

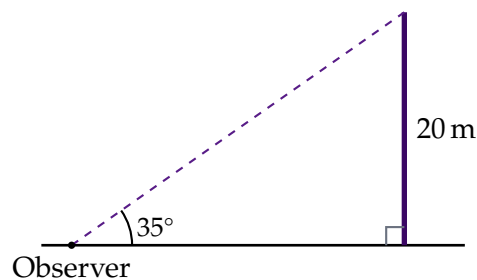
A ramp is 6 m long and rises 1.5 m.



- (a) Find the angle of the ramp above the horizontal.
- (b) Find the horizontal distance covered by the ramp.
- (c) Which trigonometric ratio gives the most direct solution?

PROBLEM 20**LEVEL II****Angle of Elevation**

From a point on the ground, the angle of elevation to the top of a 20 m tower is 35° .



- (a) Find the distance from the observer to the base of the tower.
- (b) Find the distance from the observer to the top of the tower.
- (c) Draw the triangle and identify the opposite, adjacent, and hypotenuse sides.

1.7 Vectors and Trigonometric Synthesis

PROBLEM 21

LEVEL II

Components and the Fundamental Identity

A vector has components A_x and A_y . Its magnitude is A .

Show that

$$A^2 = A_x^2 + A_y^2.$$

Then, using

$$A_x = A \cos \theta, \quad A_y = A \sin \theta,$$

recover the identity

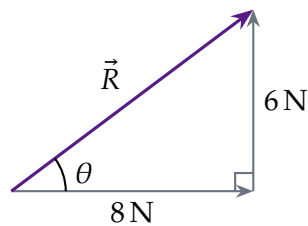
$$\sin^2 \theta + \cos^2 \theta = 1.$$

PROBLEM 22

LEVEL II

Perpendicular Forces

Two forces of 6 N and 8 N act perpendicular to each other.



- (a) Find the magnitude of their resultant.
- (b) Find the angle that the resultant makes with the 8 N force.
- (c) Explain why the Pythagorean theorem and trigonometry can be used to describe the resultant.

PROBLEM 23

LEVEL II

A Vector with Related Components

A vector has magnitude 10 and makes an angle θ with the positive x -axis.
Its x -component is twice its y -component.

- (a) Write an equation relating $\sin \theta$ and $\cos \theta$.
- (b) Determine θ .
- (c) Find both components.

PROBLEM 24

LEVEL III

Area and Hypotenuse

A right triangle has an area of 24 cm^2 and a hypotenuse of 10 cm.

- (a) Find the two possible leg lengths.
- (b) Find the two acute angles.
- (c) Explain why knowing the area and hypotenuse is enough to determine the triangle.

Hints

The hints below correspond to the problems in the handout. They are provided separately so that students can work through the problem collection without immediately seeing the associated guidance.

Hint 1 | Problem 1 – The Fundamental Identity

Consider a right triangle or the unit circle and relate the two quantities using the Pythagorean theorem.

Hint 2 | Problem 2 – A Tangent Identity

Write

$$\tan x = \frac{\sin x}{\cos x}$$

and use the fundamental identity.

Hint 3 | Problem 3 – Complementary Angles

Remember that the two acute angles of a right triangle are complementary.

Hint 4 | Problem 4 – The 3–4–5 Triangle

Identify the hypotenuse first. Then choose one of the two acute angles and identify its opposite and adjacent sides.

Hint 5 | Problem 5 – A Right Triangle with Known Sides

Use the Pythagorean theorem to find the missing side. Then identify the sides relative to each angle.

Hint 6 | Problem 5 – A Right Triangle with Known Sides

For part (e), consider what happens to

$$\tan C = \frac{\sin C}{\cos C}$$

when $C = 90^\circ$.

Hint 7 | Problem 6 – Working Backwards from Sine

Since XY is opposite the right angle, it is the hypotenuse. Use the definition of sine to determine the side opposite X .

Hint 8 | Problem 7 – The 30–60–90 Triangle

Drop an altitude in an equilateral triangle and use symmetry.

Hint 9 | Problem 8 – Exact Trigonometric Values

Use the 30° - 60° - 90° and 45° - 45° - 90° triangles.

Hint 10 | Problem 9 – The Right Isosceles Triangle

Use the Pythagorean theorem first. Then use the equality of the two acute angles.

Hint 11 | Problem 10 – Area of an Equilateral Triangle

Draw an altitude from one vertex. What kind of right triangles are formed?

Hint 12 | Problem 11 – Dividing an Equilateral Triangle

The altitude divides the equilateral triangle into two 30° - 60° - 90° triangles.

Hint 13 | Problem 12 – Similar Right Triangles

First find the missing side of ABC . Similar triangles preserve corresponding side ratios.

Hint 14 | Problem 13 – Same Angle, Different Size

The two triangles have the same value of

$$\tan \theta = \frac{\text{opposite}}{\text{adjacent}}.$$

Hint 15 | Problem 14 – Resolving a Force

Draw the force vector as the hypotenuse of a right triangle.

Hint 16 | Problem 14 – Resolving a Force

The horizontal component is adjacent to the angle, while the vertical component is opposite to it.

Hint 17 | Problem 15 – Velocity Components

Use cosine for the component adjacent to the angle and sine for the component opposite the angle.

Hint 18 | Problem 16 – Recovering an Unknown Direction

The horizontal component is adjacent to the angle, so begin with

$$F_x = F \cos \theta.$$

Hint 19 | Problem 17 – Velocity of a Ball

The 12 m/s is the magnitude of the velocity vector, not the magnitude of each component.

Hint 20 | Problem 18 – A Ladder Against a Wall

The ladder forms the hypotenuse of a right triangle.

Hint 21 | Problem 18 – A Ladder Against a Wall

For the angle between the ladder and the ground, identify the adjacent side and hypotenuse.

Hint 22 | Problem 19 – A Sloping Ramp

The ramp length is the hypotenuse, while the vertical rise is opposite the angle above the horizontal.

Hint 23 | Problem 20 – Angle of Elevation

The tower is opposite the angle of elevation, while the horizontal ground distance is adjacent to it.

Hint 24 | Problem 21 – Components and the Fundamental Identity

Substitute the expressions for A_x and A_y into the first equation.

Hint 25 | Problem 22 – Perpendicular Forces

The two forces form the perpendicular sides of a right triangle, and the resultant is its hypotenuse.

Hint 26 | Problem 23 – A Vector with Related Components

Use

$$A_x = A \cos \theta, \quad A_y = A \sin \theta.$$

Then apply the condition $A_x = 2A_y$.

Hint 27 | Problem 24 – Area and Hypotenuse

Let the two legs be a and b . The area gives one equation, while the Pythagorean theorem gives another.

Hint 28 | Problem 24 – Area and Hypotenuse

Use

$$ab = 48$$

together with

$$a^2 + b^2 = 100.$$

Look for a useful algebraic manipulation rather than immediately solving a quadratic in one variable.